

① Special Stains for Staining of Carbohydrates:

① PAS Stain (Periodic Acid Schiff Stain).
↳ Most important

② Alcian-Blue Stain → for Acidic Mucopolysaccharide

③ Combined Alcian-Blue and PAS Stain

④ Mucicarmine Stain → for epithelial Mucin.

⑤ Colloidal Iron Stain → for Acid Mucopolysaccharides

⑥ Best's Carmine Method → for Glycogen

② Special Stains for Staining of Connective Tissue:

① Collagen:-

① Masson's Trichrome Stain

② Gomori's Trichrome Stain

③ Van-Grieson Technique

③ Special Stains for Staining of Nervous Tissues:

⑨ Myelin:-

① Luxol fast Blue. (Most Common)

② Pal Weigert Method

⑩ Axons and Neurofibrils

① Bielschowsky's Silver Stain.

⑪ Astrocytes

① Cajal's Stain

④ Special Stain for Staining of Elastic Fibres

① Verhoeff's Method (Most Common)

② Resorcin-fuchsin Method

5. Special Stains for Staining of Reticulum

- ① Gordon and Sweet's Method → Most important
- ② Gomori's Method
- ③ Russell modification of Morat penta chrome stain.

6. Special Stains for Staining of lipids

- ① Oil Red O →
- ② Sudan Black B
- ③ Osmium Tetroxide Method → for unsaturated lipids.

⑦ Special Stains for Staining of bacteria:-

- ① ZN stain
- ② Gram stain

⑧ Special stains for staining of fungi:-

① Gyrocott's Methenamine Silver stain

⑨ Special stains for staining of Extracellular and Intracellular material with reference to Melanin

① Masson-Fontana stain.

② Bleaching Method.

① Special Stain for Carbohydrate

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① PAS STAIN (Periodic Acid - Schiff's Stain):-

• The PAS is widely used technique for the demonstration of Carbohydrates or glycoconjugates.

• The PAS reactive cells and tissue components include:

- ① Glycogen ② Starch ③ Mucin (Sialomucin, Neutral Mucin)
- ④ Basement Membranes ⑤ α -Antitrypsin
- ⑥ Reticulin ⑦ Fungi (Capsules) ⑧ Russell bodies.
- ⑨ Pancreatic Zymogen granules.
- ⑩ Glycolipids (Cerebrosides, gangliosides).
- ⑪ Common fungal species that are PAS reactive include:
 - Candida Albicans • Histoplasma capsulatum
 - Cryptococcus • Blastomyces.
- ⑫ Pigments (• Lipofuscin)

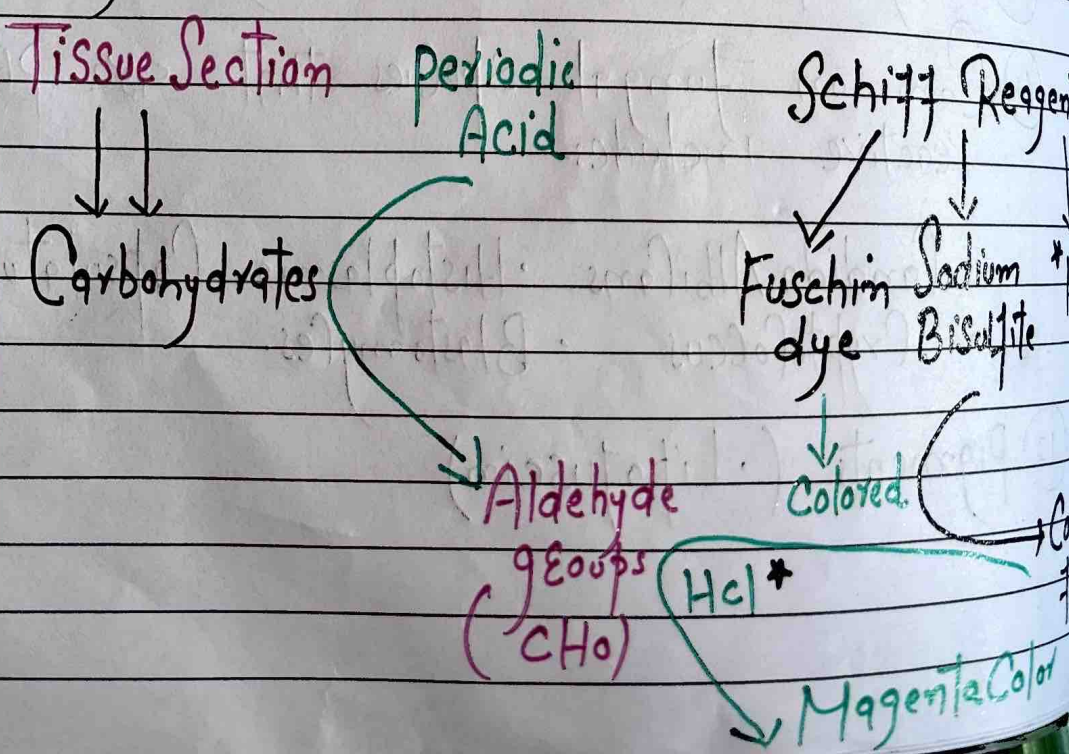
• Principle:

The PAS technique is based on the reactivity of free aldehyde groups in carbohydrates with Schiff's reagent to form a bright red / magenta end product.

The initial step in PAS technique is the oxidation of hydroxyl groups attached to adjacent carbon atoms, 1,2-glycols within the carbohydrate.

This results in the formation of two free aldehyde groups. This is produced by treatment of sections with a dilute solution of periodic acid.

These free aldehydes then react with Schiff's reagent, producing a magenta color.



Chemistry of Staining:-

• Carbohydrate in Tissue Section:-

The structure of Carbohydrate have 1,2 glycols (OH groups).

• Periodic Acid [HIO_4]

The Periodic Acid will react with OH groups of Carbohydrates and convert into aldehyde groups.

• Schiff's Reagent:-

Fuchsin
Sodium bisulfite
Acid solution (HCl)

• Fuchsin

This is a magenta color dye that combines with sodium bisulfite and becomes colorless.

• Sodium Bisulfite:-

It bleaches the color of Fuchsin and removes the original color and make it receptive to react with aldehydes.

• Acidic Solution (HCl):-

The acidic environment helps to stimulate reaction between Schiff's Reagent and

aldehydes.

Requirements:

• Periodic Acid Solution:-

- Periodic Acid = 1g (0.5-1%)
- Distilled Water = 100ml.

• Schiff Reagent:

Composition:

- Basic Fuchsin: 1g
- Sodium Metabisulfite: 1.9g
- Conc. HCl :- 100ml of 0.15M HCl
- Distilled Water:
- Activated Charcoal: 500mg

Preparation:-

- Dissolve 1g of basic Fuchsin and 1.9g of Sodium Metabisulfite ($\text{Na}_2\text{S}_2\text{O}_5$) in 100ml of 0.15M hydrochloric Acid.
- Shake the solution at intervals or on a mechanical shaker for 2 hours.
- The solution should be clear and yellow brown in color.
- Add 500mg of activated charcoal and shake for 1-2 minutes.

- Filter the Solution Via No.1 Whatman filter into a bottle.
• The filtered Solution should be clear and Colorless.
- If Solution is yellow, Repeat the Charcoal decolorisation using a fresh lot of activated Charcoal.
- Store at 4°C , Solution is stable for several months.
- Tissue Section
- Xylene (Dewax + Clearing).
- Alcohols of different grades (Rehydration / Dehydration).

Method / Procedure:-

Step 1: Dewax or Deparaffinize in Xylene for 1-2 minutes

Step 2: Rehydrate via graded alcohols [100%, 95%, 70%].

Step 3: Oxidise with Periodic Acid for 5 minutes.

Step 4: Rinse in several changes of distilled water.

Step 5:- Cover the Section with Schiff

Reagent for 15 minutes.

Step 6: Rinse in running tap-water for 5-10 minutes.

Step 7: Stain nuclei with hematoxylin. Differentiate and blue the section.

Step 8: Dehydrate in graded alcohols (70%, 95%, 100%).

Step 9:- Clear with xylene.

Step 10:- Mount with DPX and apply coverslip.

Results:

- Glycogen, neutral/Sial/Mucins :- **Magenta**
- Various Glycoproteins and PAS-reactive Substances = **Magenta**
- Nuclei := **Blue**.

NOTE:-

- Intensity of stain is dependent to some extent on length of treatment with periodic Acid and Schiff Reagent.

For basement membranes, a longer time in Periodic Acid (10mins) and Schiff Reagent (20min.) may give better results.

- Fixatives Containing Glutaraldehyde should be avoided, as it contains two aldehyde groups capable of undergoing Schiff's reaction resulting in non-specific background staining.
- Staining of glycolipids may be detected when frozen sections are used.

Control Slide:-

• Positive Control:

A tissue known to contain glycogen or basement membrane is used as positive control.

• Negative Control:-

A diastase-PAS (PAS-D) is used to confirm presence of glycogen where diastase digests glycogen leading to negative PAS staining.

Applications:

It is useful in demonstration of

- ① Glycogen (with diastase digestion)
- ② Neoplastic tumours like Carcinoma of bladder, kidney, liver, Ovary, pancreas and lungs.
Also seen in Ewing's Carcinoma, mesothelioma,
- ③ Non-neoplastic lesions like glycogen-storage disease (glycogenosis)
- ④ Connective tissue mucins in tumours
- ⑤ Mucoproteins like basement membrane of renal biopsies, thyroid Colloid, pituitary basophilic cells
- ⑥ Glycolipids like Cerebrosides, gangliosides in CNS (inborn errors of metabolism)
- ⑦ Other Carbohydrates like chitin (wall of hyaline cyst, starch (granules of skin and peritoneum)
- ⑧ Micro-organisms like fungi, amoeba.

② Special Stain for Connective Tissue

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Masson's Trichrome Stain:-

Masson's Trichrome staining is a histological staining method used to selectively stain:-

- ① Collagen
2. Collagen fibres
3. fibrin
4. Muscles
5. Erythrocytes

- This stain uses three stains for staining hence the term Trichrome is used. Therefore it is a Three Color staining protocol.
- This stain is used to distinguish Collagen from Muscles tissue specimens.
- This is also used to identify an increase in collagenous tissues.

Principle:-

Masson's trichrome is a three dye staining method:

1. Nuclear stain:-

The nuclei are first stained with iron hematoxylin.

2. Cytoplasmic stain:-

Acid dyes such as Biebrich Scarlet

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acid fuschin stains cytoplasm, Muscle
keratin, Collagen, RBCs.

These dyes are small-molecular weight
acid dyes that penetrate most tissue
elements.

3 Differentiation and Collagen staining

Treatment with phosphomolybdic
or phosphotungstic acid removes red
dye from collagen but not from muscle
cytoplasm.

Collagen has less affinity for
small acid dyes once differentiated.

- A large molecular weight dye (e.g. Aniline
or light green) then selectively stains
collagen.

Requirements:

1. Tissue Section
2. Xylene
3. Graded alcohol
4. Mounting Media.

5. Preparation of Solutions:

Solution A:

- Biebrich Scarlet : 0.9g
- Acid fuchsin : 0.1g
- Glacial acetic acid : 1ml
- Distilled water : 100ml.

Solution B:

- Phosphomolybdic acid : 2.5g
- Phosphotungstic acid : 2.5g
- Distilled water : 100ml.

Solution C:

- Aniline blue / Light green : 2.5g
- Distilled water : 100ml
- Glacial acetic acid : 1-2 ml

6. Weigert's Iron hematoxylin

7. Formal Saline

Methods / Procedure:

Step 1: Deparaffinize tissue section in xylene

Step 2: Rehydrate using 100% alcohol, 95% alcohol and 70% alcohol sequentially

Step 3: Wash in tap water

Step 4: Stain nuclei with Iron hematoxylin for 10 minutes.

Step 5: Differentiate with 1% acid alcohol

Step 6: Wash well in tap water

Step 7:- Stain in Solution A for 5 minutes

Step 8:- Rinse in distilled water

Step 9: Treat with Solution B for 5 minutes

Step 10:- Drain.

Step 11: Stain with Solution C for 2-5 minutes

Step 12: Rinse in distilled water

Step 13: Treat with 1% acetic acid for 2 minutes

Step 14: Dehydrate Via ascending grades of alcohol.

Step 15: Clear in xylene

Step 16: Mount in Permanent Mounting medium.

Results:

- Nuclei: Black / Blue
- Collagen: Green
- Muscle, Cytoplasm, Erythrocytes: Red

Applications:

- Masson's Trichrome staining technique is used for detection of Collagen fibres in tissues such as skin, heart, muscles.
- To study Muscular dystrophy and other muscular pathologies.
- To study Cardiac infections and other Cardiac pathologies.
- To study hepatic pathologies such as Cirrhosis.
- To study Kidney Pathologies such as glomerular fibrosis and interstitial fibrosis.

- To detect and analyze tumors on hepatic and kidney biopsies.
- Useful for evaluation of fibrosis in various organs like Liver, Kidney, lungs and heart.

Special Stain for Nervous Tissues.

LUXOL FAST Blue:-

This is a staining method used for staining of myelin / myelinated axons.

This stain is commonly used for identifying the basic neuronal structure in brain and spinal cord tissues.

Principle:-

The Principle of Luxol Fast Blue staining is based on selective affinity of dye to the lipoproteins present in myelin sheath of nerve fibres.

Luxol Fast Blue is a basic Copper Phthalocyanine dye which binds to the acidic groups of phospholipids and sulfatides present in myelin.

After staining, the section is differentiated using lithium carbonate and alcohol which removes excess dye from non-myelinated tissues while retaining dye in myelinated fibres.

Requirements:

1. Paraffin-Embedded tissue Section.

2. Solutions:

① Luxol Fast Blue: 0.1g
10% Acetic Acid: 0.5ml
95% alcohol: 100ml
Mix reagents and filter.

② Lithium Carbonate

③ Counterstain:

• 0.2% Cresyl fast Violet.
10% acetic acid.
Filter before use.

3. Equipment:

- Glass Slides
- Coplin Jar
- Forceps
- Microscope.

Method:

Step 1: Remove wax with Xylene.

Step 2: Take Section to 95% alcohol (not water).

Step 3: Stain in luxol fast Blue Solution for 2 hours at 60°C or 37°C Overnight.

Step 4: Wash in tap water

Step 5: Differentiate in 0.1% lithium Carbonate Solution. until gray and white matter distinguish.

• This may be more easily controlled by using 0.05% lithium Carbonate followed by 70-95% alcohol instead.

Step 6: Wash in tap water.

Step 7: Check differentiation under the microscope.

Repeat Step 5 if necessary.

Step 8: Counterstain either in 0.2% Cresyl Violet Solution or 1% neutral Red for 10 to 20 minutes.

Step 9: Drain Sections and transfer to 70% alcohol.

Avoid placing the section in water at this stage as Cresyl Violet staining loses some of its intensity.

Step 10: Dehydrate, clear in xylene and mount in DPX.

Results:

- Myelin : **Blue**
- Nuclei and Nissl Substance : **Violet / Pink**
- Erythrocytes : **deep purplish Blue**

NOTE:

- 70% alcohol differentiates Cresyl Violet stain. optimally, the Cresyl Violet should be removed, leaving cell bodies and Nissl clearly visible.
- Don't over-differentiate, the 70% alcohol will take out Cresyl Violet and to a certain extent luxol fast blue.
- Cresyl Violet counterstain will deepen color of luxol fast blue-stained myelin from turquoise to a **deep blue**.

Applications

- ① Demonstration of myelin in nervous tissue
- ② Detection of demyelination.
- ③ Diagnosis of demyelinating disease
e.g. Multiple Sclerosis
• Guillain-Barré Syndrome.
- ⑤ Assessment of extent and distribution of myelin loss
- ⑥ Demonstrate myelin damage in Conditions such as Viral encephalitis and progressive multifocal leukoencephalopathy.

Special Stain for Elastic Fibres

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Verhoeff's Elastic Fibre Stain

Verhoeff's method is the Classical method for elastic fibres and works well for all routine fixatives.

It is a histological staining procedure developed by Frederick H. Verhoeff in 1908.

It is one of the most commonly used stains to visualize elastic tissues as found in:-

- Blood Vessel walls
- Elastic Cartilage
- Lungs
- Skin
- Bladder and some ligaments.

Principle:-

The Principle is based on the Regressive Staining using an iron-hematoxylin lake.

1. Mordanting and Oxidation

The tissue is first Overstained using a working solution of hematoxylin, ferric chloride and Iodine.

ferric chloride and Iodine serve a dual purpose. They act as

Mordant:- link dye to tissue

Oxidizer:- to Convert hematoxylin into its active form hematein

The resulting iron-hematein complex has high affinity for elastic fibres, by binding via hydrogen and Vanderwaals forces.

2. Regressive Differentiation:-

The section is treated with the excess ferric chloride. This act as differentiator by breaking the dye-mordant bond in most tissues making them colorless.

The elastic fibres have strongest bond with dye. So they retain the black stain.

3. Iodine Removal:-

Sodium Thiosulfate is used to remove any residual iodine.

4. Counterstaining:-

The procedure is typically followed by Van Gieson Counterstain (Acid fuchsin and Picric Acid).

The Picric acid acts as secondary differentiator, further refining black elastic fibres against a yellow (cytoplasm) and red

Collagen.

1. Overstain
with a
Hematoxylin -
Ferric Chloride
Iodine Solution

FE chloride +
Iodine acts as

Mordant Oxidizer

→ Differentiation
in Ferric Chloride
which breaks the
Mordant-Dye Complex and
Elastic fibres will break

↓
Remove Excess FeCl
and Iodine with
ThioSulfate.

Counterstain in
Van Gieson Consisting
of Picric Acid AND
Acid fuchsin

Requirements:-

1. Tissue Section
2. Xylene
3. Graded alcohol
4. Coverslips.

• Preparation of Stain:-

Solution A:-

Hematoxylin: 5g
Absolute alcohol: 100ml.

Solution B:-

Ferric chloride:- 10g
Distilled Water: 100ml

Solution C:-

Iodine:- 1g
Potassium Iodide:- 2g
Distilled Water:- 100ml

• Verhoeff's Solution:

Solution A:- 20ml

Solution B:- 80ml

Solution C:- 80ml.

• Von Gieson stain:-

- Picric Acid
- Acid Fuchsin

Method:-

Step 1: Deparaffinize tissue section in xylene

Step 2:- Rehydrate in descending grades of alcohol.

Step 3:- Stain in Verhoeff's Solution for 15-30 minutes or until section is black.

Step 4:- Rinse in water

Step 5:- Differentiate in 2% aqueous ferric chloride until elastic tissues appear black on a gray background.

If section is overdifferentiated, it may be returned to Verhoeff's Solution.

Step 6:- Rinse in water and then transfer to 95% alcohol or Sodium Thiosulfate to remove Iodine.

Step 7:- Counterstain as desired. Van Gieson is conventional, although Eosin may be used.

Step 8:- Blot to remove excess stain

Step 9: Dehydrate rapidly Through ascending grades of alcohol.

Step 10: Clear in Xylene and Mount in Permanent Mounting Media.

Results:

1. Elastic fibres : Black.
2. Nuclei : Blue to Black
3. Collagen : Red
4. Other tissue elements (Muscle fibres, RBCs) : Yellow.

Applications:

- ① Demonstration of elastic fibres
- ② Study of blood Vessels (helps in identifying elastic laminae of arteries)
- ③ Diagnosis of Vascular disease
e.g. Atherosclerosis, Vasculitis where fibres show fragmentation or loss.
- ④ Demonstration of elastic fibres in alveolar walls useful in emphysema and other chronic lung disease.
- ⑤ Study Tumors rich in elastic fibres such as elastofibroma.

Special stain for Reticulin fibres:-

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Gordon and Sweet's Methods:-

Gordon and Sweet's Reticulin Stain:-

It is a Silver impregnation technique used for demonstration of Reticular fibres (Reticulin)

Principle:

The Principle of Gordon and Sweet's Reticulin Stain is an oxyphilic Silver impregnation process.

• Since Reticulin fibres (Type III Collagen) have no natural affinity for Silver, they must be chemically "primed" to capture and eventually reduced Silver into a visible black deposit.

Steps by step Chemical Principle:-

1. Oxidation:-

Potassium Permanganate oxidises Carbohydrate (glycoprotein) chain in Reticulin fibres.

This process generates reactive aldehyde groups that serves as initial attachment points for the stain.

2. Bleaching:- (cleaning)

Oxalic Acid is used to remove purple tint of permanganate, leaving the tissue colorless.

3. Sensitization:-

- The tissue is treated with sensitizer such as iron alum. This metal salt binds to reactive sites on reticulin fibres, acting as a bridge to facilitate later deposition of silver.

4. Silver impregnation:-

The sections are treated with an ammoniacal silver solution (aka diamine silver).

During this step, silver ions replace iron alum sensitizer and bind to reticulin fibres.

5. Reduction:-

A reducing agent, usually 10% formalin converts colorless silver ions into a visible black metallic silver precipitate.

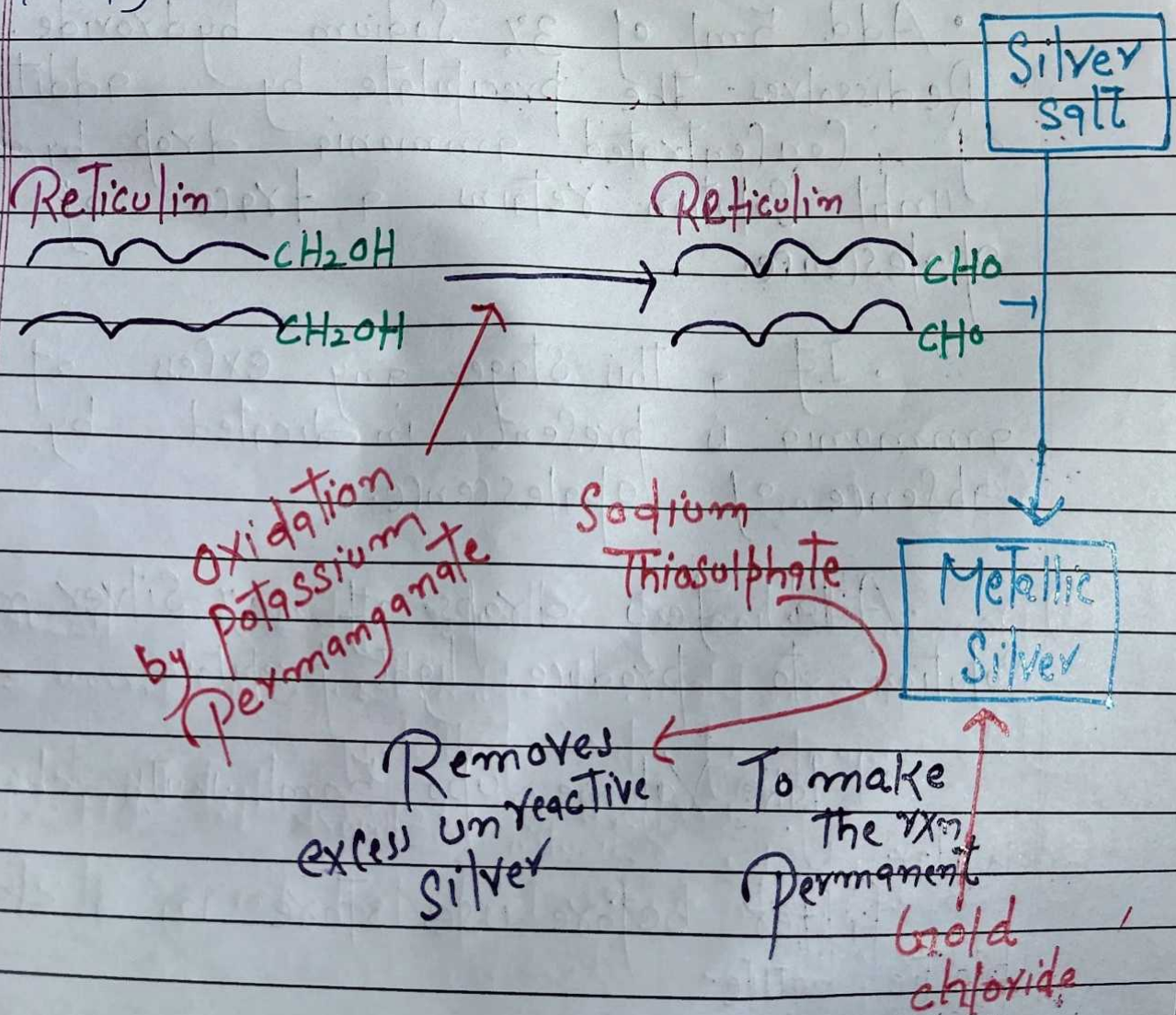
6. Toning:- Gold chloride replace the unstable

Silver with metallic gold. This provides chemical stability (prevent fading) and shifts color from brownish-black to deep, permanent jet black.

7. Fixation:

Sodium thiosulfate dissolves and removes any unreacted excess silver salts.

This prevents the background from darkening or pepper like artifacts from forming when slide is exposed to light.



Requirements:

1. Tissue Section
2. Xylene
3. Graded glass
4. Mounting Media
5. Coverslips

• Preparation of Silver Solution:

- Take 5ml of 10% aqueous Silver nitrate Solution add Concentrated ammonia drop by drop, until precipitate first formed dissolves, taking care to avoid any excess ammonia.

- Add 5ml of 3% Sodium hydroxide.
- Redissolves the precipitate by addition of Concentrated ammonia drop by drop until solution retains a trace of opalescence.

- If at this stage any excess of ammonia is present, indicated by absence of opalescence.

- Add few drops of 10% Silver nitrate Solution to produce light precipitate.

- Make Volume up to 50ml with distilled water.

filter before use. Store in a dark plastic bottle.

Method:-

Step 1: Deparaffinize sections and take to water

Step 2: Oxidise by treating with 1% Potassium permanganate solution.

Step 3: Rinse in tap water

Step 4: Bleach with 1% oxalic acid solution

Step 5:- Rinse in tap water

Step 6:- Treat with 2.5% iron alum solution for at least 15 minutes.

Step 7:- Wash well in several changes of distilled water.

Step 8: Place in Coplin jar of Silver solution for 2 minutes

Step 9: Rinse in several changes of distilled water.

Step 10: Reduce in 10% aqueous formalin solution for 2 minutes

Step 11: Rinse in tap water

Step 12: Tone in 0.2% gold chloride solution

for 3 minutes.

Step 13: Rinse in tap water

Step 14: Treat with 5% Sodium Thio solution for 3 minutes

Step 15: Rinse in tap water

Step 16: Counterstain as desired

Step 17: Dehydrate via ascending grades of alcohol.

Step 18: Clear in xylene and Mount in Permanent mounting Medium.

Results:

Reticular fibres: Black
Nuclei: Red.

Applications:

- ① Demonstration of Reticular fibres in tissues liver, Spleen, lymph nodes, Bone Marrow etc
- ② Assessment of liver pathology e.g. Cirrhosis (Thickened, distorted, collapsed)
- ③ Demonstrates reticulin network in BM, spleen and lymph nodes. Important in evaluating myelofibrosis, leukemia, lymphoma's.

Special stain for Lipids

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Oil Red O :- (Lipid stain) :-

This is used for staining the lipids on frozen sections, fresh smears or touch preps.

It is used in the histological visualization of fat cells and neutral fats.

- fat is alcohol soluble, section from paraffin embedded blocks cannot be used. Cryostat sections of fresh unfixed tissue are used for staining of fat.

Principle:-

The principle of Oil Red O staining is based on physical process of selective solubility known as solvent partitioning.

Unlike most histology stains that rely on chemical bonds, ORO is a lysachrome (fat soluble dye) that colors lipids by moving between different solvent phases.

Mechanism:-

1. Lipid Solubility:- ORO is a lipophilic dye that readily dissolves in neutral fats.

2. Solvent Migration:-

The ORO is initially dissolved in Organic Solvent e.g. isopropanol but its solubility in this decreases significantly when dilute with water or when it encounters tissue.

3. Partitioning:-

During staining, the ORO molecules move from surrounding solvent into the lipid droplets within tissues because they are much more soluble in lipids.

4. Coloration:-

This transfer results in accumulation of red dye within fat, making lipid deposits visible as **bright red** structures under microscope.

Requirements:-

1. ~~Frozen~~ Section
2. Microscope
3. Coverslips
4. Mounting Medium

Solutions:

1. Stock Solution:-

Saturated Solution of Oil red O
(300mg/dl) in 99% isopropanol.

2. Working Solution:-

- Mix Six parts of stock oil red O with four parts of distilled water.

- Allow to stand for 10 minutes and then filter.

This Solution should be used within 2-4 hours.

Keep in air tight staining Vessels to prevent precipitation of dye as a result of evaporation.

3. Mayer's or Ehrlich's or Harris hematoxylin

4. 0.05% Sodium or Lithium Carbonate.

Procedure:-

Step 1:- Take a frozen tissue Section.

Step 2: Rinse Section well in distilled water

Step 3: Stain in Oil Red O for 10 minutes

Step 4: Rinse in tap water

Step 5: Counterstain in Mayer's hematoxylin for 2-3 minutes or Harris's hematoxylin for 1 minute

Step 6: Rinse in tap water

Step 7: Blue in 0.5% Lithium Carbonate

Step 8: Rinse well in several changes of distilled water

Step 9: Mount in glycerine jelly or any good aqueous mounting Medium.

Results:-

Lipids (Neutral fats) :- **Red**
Nuclei :- **Blue**

Other tissues: Pale brown.

Applications:

- ① Demonstration of fats in fatty liver
- ② Diagnosis of lipid storage disease
e.g. Gaucher disease.
- ③ Demonstration of fat embolism.
Fat droplets in lungs or tissue.
- ④ Detection of lipids in atherosclerotic plaque.
- ⑤ Demonstration of lipid in muscle biopsies.

Special Stain for bacteria

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ZN Stain [Ziehl-Neelsen Stain]:-

Zn Stain (Acid fast stain) is a differential stain which divided micro-organisms into two groups:-

Acid fast and
Non-Acid fast.

This stain was initially introduced by Paul Ehrlich and subsequently modified by German bacteriologists Franz Ziehl and Friedrich Neelsen.

This technique is used on micro-organisms that are not easily stained by basic stain such as Gram stain.

One of the most complex microorganism that require harsh treatment of ZN compound is Mycobacterium.

- Mycobacterium, Actinomyces, Nocardia, isopora, Cryptosporidium and some fungi contain thick wall made up of lipoidal complexes known as mycolic acid.

• Mycolic acid is difficult to stain so strong staining solutions containing phenol preferably with heat are used to

Stain them. Once stained they
decolourisation with mineral
hence called Acid fast bacilli.

Resist
acids

Principle:

- The Principle of ZN stain is based on the high lipid content of mycolic acid present in cell wall of acid-fast bacilli.

This waxy substance make organisms resistant to decolorisation by strong mineral acids.

- During staining, tissue section is treated with Carbol Fuchsin (Basic Fuchsin Phenol) and heat.

- The application of heat increases permeability of cell wall, allowing dye to penetrate and bind firmly to cellular component.

- On subsequent treatment with an acid-alcohol decolorising agent, acid fast bacilli retain 1st stain, where as non-acid fast organisms are decolorised.

A Counterstain (Methylene Blue / Malschite green) stains background and non acid fast organisms.

Requirements:

• Specimen: Tissue Section

Equipment:

- Bunsen Burner
- Staining Rack
- Microscope.

Solutions:

① Carbol fuchsin:

• Basic fuchsin: 1g

• Absolute ethyl Alcohol: 10ml

• Phenol Crystals: 5g

• Distilled water: 100ml

Dissolve basic fuchsin in ethyl alcohol and phenol in water.

• Mix two solutions and filter before use.

② Acid alcohol:

1% HCl in 70% Alcohol.

③ Counterstain:

1% aqueous solution of Methylene Blue or
0.5% aqueous solution of Maltichite Green.

Procedure:

step 1: Bring Tissue Section to water

step 2: Flood stain with Carbol Fuchsin
Heat from underneath until steam rises.

Reheat at three times interval for five minutes.

(OR)

Stain in jar at 37°C for 1 hour.

step 3: Wash well in tap water

step 4: Decolorise with 1% acid alcohol until a faint pink colour is retained

step 5: Wash in water

step 6: Counterstain with The 1% Methylene Blue or 0.5% Malachite green for 20-30 seconds.

step 7: Wash in water.

step 8: Dehydrate, clear and Mount with DPX.

Results:

Acid fast bacilli = Red

cells and Nuclei = Blue/Green

Other bacteria = Blue/Green

RBCs = Slightly reddish tint.

Applications:

① Demonstration of Acid fast bacilli Such as:
M. Tuberculosis
M. Leprae.

② Detection of other Acid fast organisms.
Mycobacteria
• Cryptosporidium
• Isospora, Cyclospora in intestinal biopsies

③ Help in identification of Acid-fast bacilli in tissue sections of
a) Tuberculous granuloma
• Leprosy lesions
• Pulmonary or extrapulmonary Tuberculosis.

Special Stain for Fungi

Methenamine Silver Stain (Grocott's)

Modification of Grocott's Methenamine Silver Method:-

It is a Valuable method Used for demonstration of Fungi in tissues.

Principle:-

The Cell wall of Fungi contain polySaccharides (Chitin, glucan). These PolySaccharides are oxidised

by Chromic acid to produce aldehyde groups (CHO).

The Section is then treated with methenamine Silver nitrate solution.

The aldehyde groups reduce the methenamine Silver nitrate in alkaline solution to metallic Silver

The reduced metallic Silver deposits appear black or dark brown at sites where fungal cell wall is present.

• The background tissue is green Counterstained by light green for Contrast.

D.

Borax (Sodium borate solution) acts as buffer
maintain alkaline pH.

1. Oxidise in
chromic acid
To produce
Aldehyde groups

→ Remove Residual
chromic acid
with Na
Bisulfite

Counter Stain
(light green)

↓
Silver Nitrate
Silver ions bind &
reduce to metallic
Silver

↑
Unreduced
Silver is
Removed via
Sodium Thiosulfate

↓
Tone in Gold
Chloride
from Silver to
metallic solution.
Gold, More
clarity, contrast
& stability

Requirements:

- Tissue Section
- xylene
- Microscope.
- Graded alcohol.
- Mounting Medium

Solutions:

① 5% Chromic Acid:-

Chromium Trioxide = 5g
Distilled Water = 100ml

② Stock Solution of Methenamine Silver nitrate:-

- ① 5% borax (sodium borate) in distilled water
- ② 5% Silver nitrate in distilled water 5ml
- ③ 3% Methenamine in distilled water 100ml

Shake to dissolve a white precipitate that forms. Filter the solution into a brown bottle. The clear solution can be kept for several months when refrigerated.

3. Working Solution of Silver nitrate

5% borax solution = 2ml
Distilled water = 25ml.

Mix. Add 25ml of Methenamine Silver into stock solution.
filter into chemically clean container.

4. 1% aqueous Sodium bisulphite.

5. 1% Gold Chloride:- 10ml
Distilled water = 90ml.

6. 2% Sodium Thiosulphate.

7. Light Green Solution.

Light green SF: 0.2g

Glacial acetic acid: 0.2ml

Distilled water: 500ml.

Mix well and filter.

Procedure:

Step 1: Bring tissue section to distilled water

Step 2: Oxidise in 5% Chromic acid for 1 hour.

Step 3: Wash in running tap water for few seconds.

Step 4: Rinse in 1% Sodium bisulphite for 1 minute to remove excess residual chromic acid.

Step 5: Wash in tap water for 10 minutes.

Step 6: Rinse in four changes of distilled water.

Step 7: Place in freshly prepared methenamine Silver nitrate Working Solution at 58°C for 40-60 minutes, or until section is yellow brown.

- Using paraffin Coated forceps remove slide and rinse in warm distilled water. and check microscopically for adequate Silver impregnation.

Fungi should stain dark brown. Rinse in distilled water and return to Silver Solution if section is too pale.

Step 8: Rinse in six changes of distilled water

Step 9: Tone in 1% gold Chloride for 2-5 minutes.

Step 10: Rinse in distilled water.

Step 11: Treat with 2% Sodium Thiosulphate Solution for 2-5 minutes to remove Unreduced Silver

Step 12: Wash thoroughly in tap water for five minutes.

Step 13: Counterstain with light green solution for Ziehl-Neelsen.

Step 14: Dehydrate, clear in xylene and mount in a synthetic resin medium.

Results:

Fungi:- Sharply outlined in black
Mucin:- grey.

Background:- pale green

Applications:

- ① fungal detection:
Useful for: Candida, Aspergillus, Histoplasma, Cryptococcus etc.
- ② Used in lung biopsies for detection of pneumocystis pneumonia, especially in immunocompromised patients.
- ③ Lungs, Skin, Gastrointestinal tissue sections can be studied for fungal invasion.

special stain for Extracellular and intracellular material with reference To Melanin.

Masson-Fontana Method: (Masson-Fontana Ammoniacal Silver Reaction):

Masson-Fontana stain is a special histochemical Silver impregnation method used for demonstration of melanin pigment (whether present intracellularly and extracellularly) and other argentaffin substances in tissue sections.

Principle: The Masson-Fontana method is based on argentaffin property of melanin.

Melanin present in tissue sections has ability to reduce ammoniacal silver nitrate solution to metallic silver without aid of external reducing agent.

As a result, Melanin granules whether present intracellularly or extracellularly are stained black, while background tissue is demonstrated by suitable counterstain.

Requirements:

- Tissue Section

① Preparation of Silver Solution:

- Add Strong (28%) ammonia water drop by drop to 20ml of 10% aqueous silver nitrate until dark brown precipitate that form is almost dissolved. (about 2ml ammonia water will do it)

Add 20ml Distilled water and leave to stand for 10-24 hours.

Place the Supernatant in a dark brown bottle and will keep for weeks. Filter before use.

② 0.1% gold chloride

③ Sodium Thiosulphate Solution (5%)

④ Counterstain:

Saffranin | Neutral Red.

Procedure:

Step 1: Bring the Section to distilled water.

Step 2: Treat with the ammoniacal Silver solution in a Coplin Jar which has been covered with aluminium foil for 30-40 minutes at 56°C or overnight at room temperature.

Step 3: Wash well in several changes of distilled water.

Step 4: Tone in 0.1% gold Chloride for 3-5 minutes.

Step 5: Rinse in distilled water.

Step 6: Treat Section with 5% aqueous Sodium Thiosulfate (hypo) for 1 minute.

Step 7: Wash well in running tap water for 2-3 minutes.

Step 8: Counterstain with 0.5% Safranin or 1% Neutral red for 1-2 minutes.

Step 9: Rinse in distilled water.

Step 10: Dehydrate, Clear and mount in a Synthetic Resin.

Results:

- Melanin : Black
- Argentaffin : Black
- Nuclei : Red

Applications:

- ① Demonstration of melanin:-
Intracellular Melanin = Seen in melanocytes, malignant melanoma, Nevus cells.
Extracellular Melanin =
Seen in areas of melanin-leakage after cell damage or dermal melanin deposition.
- ② Identification of melanocytic lesions:-
Used in diagnosis of benign and malignant melanocytic tumours.
- ③ Differentiate melanin-containing cells from other pigments.
- ④ Study of pigment distribution in skin, eyes.